MARVELL TECHNOLOGY, INC.

Nasdaq: MRVL | Optical DSPs, interconnect, switching and custom silicon for AI data centres | Santa Clara, California

Tommy Lau | 8 October 2026 | Initiation of coverage | Independent research

INITIATE AT LONG	**Target price:** US$355.00	**Last price:** US$284.68 (Nasdaq close, 7 October 2026)	**Upside:** +25%

Marvell's optical toll is still the biggest line in its own plan. At US$285, the price needs little growth after fiscal 2028.

Marvell makes the optical DSPs that turn AI network traffic into light and back, plus the switch chips and custom chips around them. On 28 September I argued that Marvell is paid every time AI chips talk, whoever made the chips. The results support it: data centre revenue rose 46% in the quarter to 1 August 2026, to 79% of the total, and Marvell guides interconnect, its largest data centre business, to grow more than 70% this fiscal year.

At its Investor Day on 6 October Marvell raised its fiscal 2028 outlook to about US$20bn, from US$18bn, and set a fiscal 2031 target of US$70bn to 90bn. At the midpoint interconnect is about US$37.5bn of it, the largest line, ahead of custom chips at about US$30bn. At US$284.68 and 33 times, the price needs fiscal 2029 revenue of about US$22.5bn, only 12% above the new fiscal 2028 outlook.

My call: LONG, medium conviction, target US$355. My base case takes US$28bn for fiscal 2029, below the path management set out, and is worth US$356, up 25%. The bear case is US$153 (down 46%) and the bull case US$517 (up 81%).

Stock data
LAST PRICE **US$284.68, Nasdaq close 7 October 2026; 52-week range US$70.69 to 329.88 (intraday)**
MARKET VALUE **About US$250bn on 876.9m common shares; US$256bn with Nvidia's preferred as converted**
NET DEBT, 1 AUG 2026 **US$1.03bn: debt US$4.96bn, cash US$3.93bn**
NON-GAAP P/E, MY ESTIMATES **68x FY27E, 38x FY28E, 26x FY29E (years to about January)**
CONSENSUS (STOCKANALYSIS.COM) **EPS US$4.22 FY27, US$6.75 FY28; average target US$335.83**
Q3 FY27 GUIDE (27 AUG) **Revenue US$3.15bn +/- 5%; non-GAAP gross margin 57.5 to 58.5%; EPS US$1.05 to 1.15**
INVESTOR DAY (6 OCT) **Revenue about US$20bn in FY28 (was US$18bn); custom US$12bn+ in FY29; US$70bn to 90bn in FY31**
FY31 TARGET MODEL **Gross margin 56 to 59%; operating margin 44 to 46%; tax 15%; FCF above 36%; EPS above US$30**
CUSTOMERS, FY26 **Ten largest took 82% of revenue (81% in FY25)**
NEXT EVENT **Q3 FY27 results expected early December 2026; date not yet announced**

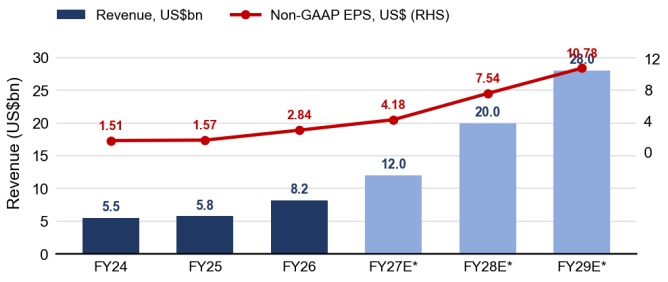
Key financials (non-GAAP unless marked; fiscal years to about 31 January)

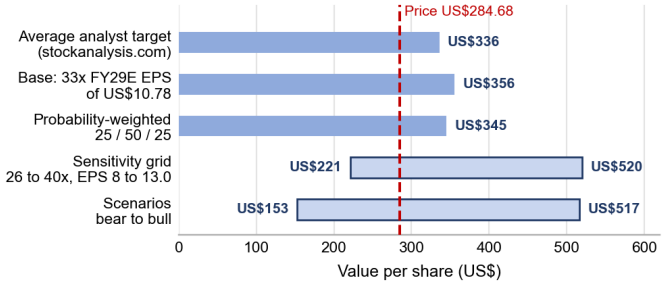
US$ billion unless stated	FY24A	FY25A	FY26A	Q3 FY27 guide	FY27E*	FY28E*	FY29E*
Revenue	5.51	5.77	8.19	2.99 to 3.31	12.00	20.00	28.00
Revenue growth (%)		5	42		46	67	40
Data centre revenue	2.22	4.16	6.10		9.75	18.00	
Gross margin (%)	61.2%	61.0%	59.5%	57.5 to 58.5	58.4%	58.0%	
Operating income	1.60	1.67	2.89		4.46	8.20	12.04
Operating margin (%)	29.0%	28.9%	35.3%		37.1%	41.0%	43.0%
EPS, diluted (US$)	1.51	1.57	2.84	1.05 to 1.15	4.18	7.54	10.78

Consensus EPS (US$)		4.22	6.75	
P/E at US$284.68 (x)	100	68	38	26

**FY27E to FY29E are my own estimates; FY28E revenue is management's US$20bn outlook of 6 October 2026. FY2027 is the year to about 30 January 2027. Income lines are Marvell's non-GAAP measures, which exclude stock-based pay (US$326m in Q2 FY27, 11.9% of revenue) and acquisition amortisation; GAAP EPS in the first half of FY27 was US$0.38 against US$1.75 non-GAAP. Operating income to FY26 is revenue times the reported margin. Consensus from stockanalysis.com: FY27 updated 7 October 2026; FY28 from S&P Global data of 2 October, before the Investor Day. History from Marvell's results releases.*

Two charts: earnings that compound, and a value above the price



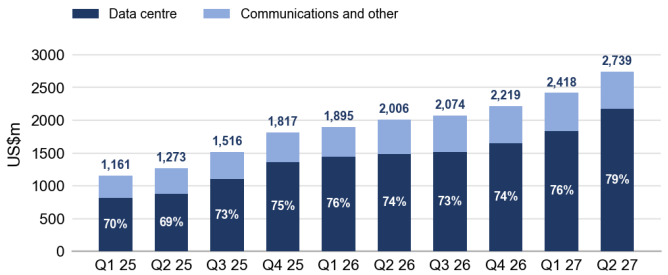


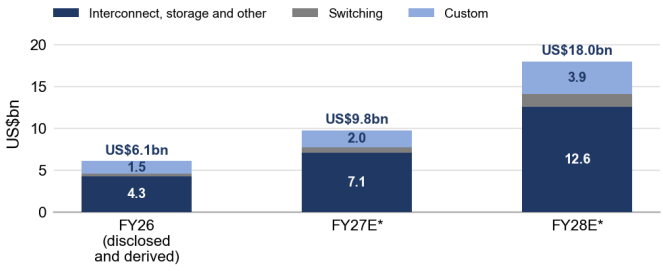
Left: revenue and non-GAAP diluted EPS, FY24A to FY29E; FY24A to FY26A per Marvell, the rest my estimates. Right: value per share from the base case, the probability-weighted cases, the sensitivity grid and the bear to bull range, against the US$284.68 price (red) and the stockanalysis.com average analyst target, updated 7 October 2026.

The thesis: the translators at each end of the torch

Go back to the torches from the September piece. Shout to someone three streets away and the words smear into noise. Flashes of light carry, but someone at each end must turn words into flashes and back again, without a mistake. Those two translators do not care who is talking. More talkers, talking faster, need more of them.

Marvell is one of those translators. It bought the job in April 2021, when it completed its purchase of Inphi. Its optical DSPs sit inside the transceivers at each end of a fibre, cleaning up a signal that carries two bits per pulse. Around them it sells the amplifiers and drivers beside the laser, the modules that link data centres, and the Ethernet switch chips the plugs slot into. Data centre revenue was US$2.17bn in the quarter to 1 August 2026, 79% of the total and up 46% on a year earlier. In May Marvell guided interconnect, the largest part of the data centre business, to grow more than 70% in fiscal 2027; in March it had guided custom chips to grow more than 20%, from US$1.5bn in fiscal 2026.





Left: revenue by fiscal quarter, data centre (with its share) and communications and other, US$m; Q2 FY27 is the quarter to 1 August 2026. Source: Marvell results releases, 3 December 2024 to 27 August 2026. Right: data centre revenue by business. FY26 custom (US$1.5bn) and switching (above US$0.3bn) as disclosed on the 5 March 2026 call, the remainder derived; FY27E and FY28E are my split around management's outlook. Marvell does not report interconnect in dollars.

The test the piece set. It said to watch how much growth comes from optics rather than custom chips. On my split, the remainder outside custom and switching, mostly interconnect, grows 65% in fiscal 2027 and supplies about 77% of the data centre growth; custom supplies about 14%. In fiscal 2028 data centre revenue rises to about US$18bn, Marvell's Investor Day figure, and on my split the remainder still supplies about 67% of the growth, custom about 23%. Custom's turn comes in fiscal 2029: Marvell now targets more than US$12bn of custom revenue that year, more than three times fiscal 2028.

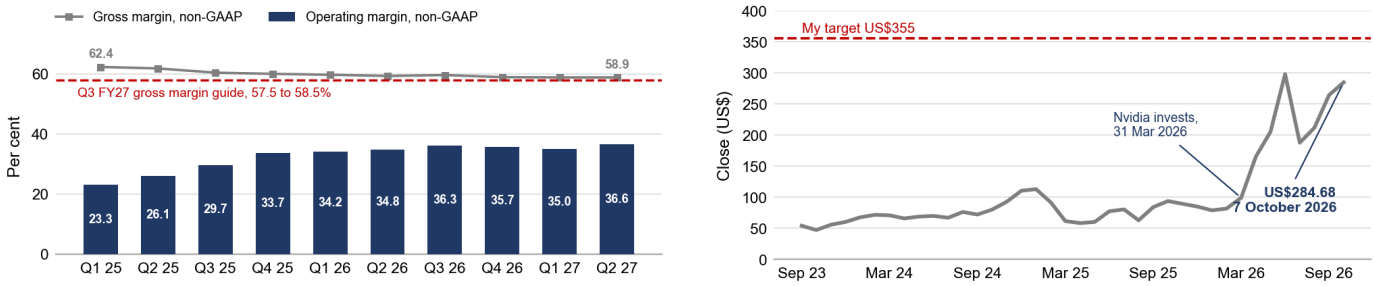
The prepayments buy capacity for the whole business

Marvell plans about US$1bn of capacity prepayments to suppliers in fiscal 2027. Dan Durn, the finance chief, gave the plan on the May call and repeated it on the call of 27 August. Neither call tied the money to optical DSPs or switch chips. Durn said the prepayments back the design wins Marvell has secured and will be set against future material purchases.

The 10-Q for the quarter to 1 August describes "capacity reservation arrangements with certain foundries and partners", and the 10-K's commitments note names foundries and test and assembly partners. The balance sheet line, prepayments on supply capacity reservation agreements, rose from US$278.8m to US$487.0m in six months. So the prepayments show a company buying capacity ahead of demand across the whole business, in a year when custom chips ramp hard in the second half. They are not evidence for the optical half in particular. The claim does not rest on the prepayments, and the revenue mix still supports it.

The plug is staying, for now

My piece's test was whether the translation moved off the plug, into linear or co-packaged optics. On the August call Matt Murphy, the chief executive, said pluggable modules remain the main form factor for scale-out networks and that he did not expect that to change; 800G demand remains strong and 1.6T is ramping rapidly. The twist is scale-up, the links inside one large AI system, where Marvell expects optics to move close to the chip. It bought Celestial AI for US$3.5bn in February to make co-packaged parts and forecasts a US$500m annualised run rate from them in the fourth quarter of fiscal 2028. At the Investor Day it put interconnect revenue growth at about 65% a year from fiscal 2026 to 2031, behind only custom at about 80%. Where the translation does leave the plug, Marvell is so far one of the companies taking it.



Left: non-GAAP operating margin (bars) and gross margin (line) by fiscal quarter, against the Q3 FY27 gross margin guide; gross margin slips as custom grows, operating margin rises on cost control. Right: MRVL month-end close, September 2023 to the 7 October 2026 close (Nasdaq.com to 5 October, stockanalysis.com after), against my target.

Variant perception: what the market prices in, and where I differ

At US$284.68 Marvell is worth about US$250bn on common shares and US$256bn with Nvidia's preferred as converted, with US$1.0bn more debt than cash. The shares closed at US$73.73 on 4 February 2026, peaked at a close of US$316.43 on 4 June (intraday high US$329.88 on 18 June), and fell to US$183.30 on 30 July. They rose 5.8% on the day of the Investor Day, to US$287.01, and have risen 235% since the end of 2025 and 8.7% since my piece. They trade at 100 times FY26 non-GAAP EPS of US$2.84, and 38 times my FY28 estimate.

What the price needs. By October 2027 the market will price fiscal 2029. At 33 times, the median of six peers, US$284.68 needs FY29 earnings of US$8.63 a share. At my 43% operating margin that needs revenue of about US$22.5bn: management's US$20bn for FY28, already up 67%, plus another 12%. Marvell's custom target alone adds about US$8bn in FY29 on my FY28 custom estimate, so the price allows for most of it to slip.

Where I differ. I do not take management's path. Getting from US$20bn to the US$80bn midpoint in three years means growing about 59% a year, and I take 40% for FY29. Even so, the price needs less than I assume. The growth beyond FY28 leans towards custom, and the Google deal points that way: its warrant for 59.0m shares at US$206.58 vests one tranche for each US$500m of custom revenue, across 240 tranches, so it covers up to US$120bn through fiscal 2033. Custom work is lumpier and the buyers are stronger. Non-GAAP gross margin has slipped from 62.4% in the quarter to May 2024 to 58.9%, and Marvell guides 57.5 to 58.5% this quarter because of the custom ramp; the FY31 model allows 56 to 59%, "influenced by product mix". So I pay no more than the peer median multiple for it, and get the upside from earnings, not from a rerating. The toll stays the largest line: interconnect is 47% of the FY31 midpoint.

Valuation, with the working

I value Marvell on non-GAAP EPS, the basis on which it guides, twelve months out, so on FY29. FY27 takes the reported first half, the third quarter at the guide's midpoints (US$3.15bn, 58% gross margin, US$655m of costs; EPS US$1.10) and a fourth quarter of US$3.69bn, what "roughly $12 billion" leaves, at 38.8% operating margin. FY28 takes management's US$20bn, 58% gross margin, costs growing at about half the rate of revenue, as the Investor Day model repeats, and the guided 13% tax: 41.0% operating margin and US$7.54 a share, against a pre-Investor Day consensus of US$6.75. FY29 grows 40% at

43% operating margin, on the way to the 44 to 46% target, with the long-term 15% tax rate and 940m shares. 33 x US$10.78 = US$356; my target is US$355.

Scenarios: three cases, twelve months out

Case	What happens	FY29 EPS	Multiple	Value	Change	Weight
Bear	AI spending pauses: no growth in FY29, operating margin 39%	US$6.94	22x	US$153	down 46%	25%
Base	Growth 40%, operating margin 43%	US$10.78	33x	US$356	up 25%	50%
Bull	Management's path: growth 60%, operating margin 45%	US$12.91	40x	US$517	up 81%	25%
Weighted	**25 / 50 / 25**			**about US$345**	**up 21%**	

All cases start from management's FY28 outlook of US$20bn. The bear case is not fanciful: the shares closed at US$58.37 at the end of April 2025, and revenue fell in fiscal 2024. The band: I go long when my base value is 15% or more above the price; at US$356 it is 25% above.

Sensitivity: FY29 EPS against the multiple

FY29 EPS	26x	33x	40x
US$8.50	US$221	US$281	US$340
US$10.78	US$280	**US$356**	US$431
US$13.00	US$338	US$429	US$520

Six of the nine cells sit above today's price, and six by more than 15%. Each of the three at or below it needs a multiple of 26 times, or FY29 earnings of US$8.50 at 33 times. Every figure is live in the accompanying model, so a change to growth, margin, tax, shares or the multiple moves the value.

Peer landscape

Company	Ticker	Forward P/E	What they make
Marvell	Nasdaq: MRVL	50.4x	Optical DSPs, TIAs and drivers, DCI modules, Ethernet switch chips, custom AI chips
Astera Labs	Nasdaq: ALAB	69.9x	Retimers, PCIe and scale-up switches, smart cable modules
Lumentum	Nasdaq: LITE	51.5x	Indium phosphide lasers, transceivers, optical circuit switches
Coherent	NYSE: COHR	36.0x	Lasers, optical transceivers, materials
Credo	Nasdaq: CRDO	30.8x	Active electrical cables, SerDes and optical DSPs
Broadcom	Nasdaq: AVGO	20.9x	Switch chips, optical DSPs, custom AI chips
Nvidia	Nasdaq: NVDA	19.8x	GPUs, NVLink, InfiniBand and Ethernet networking

Forward P/E from stockanalysis.com, retrieved 6 October 2026 during US trading; Marvell's at the 7 October close. The peer median, excluding Marvell, is 33.4 times. Multiples only; no view on the peers' shares.

Shares and dilution

Marvell had 876.9m common shares on 21 August 2026 and guides to 921m diluted this quarter. On top sit Nvidia's US$2.0bn of preferred stock, bought on 31 March, which converts into 21.8m shares at US$91.84 and is worth about US$6.2bn at the price; the Google warrant, which would add about 16m net shares at the price if fully vested; and up to 22.4m shares and about US$233m in cash still due to Celestial AI's sellers on revenue milestones through fiscal 2029. Stock-based pay was US$326m in the August quarter, 2.1 times a year earlier. Buybacks, US$400m in the first half, offset some of it. I use 930m diluted shares for FY28 and 940m for FY29.

Catalysts

When	Event	What I watch
Early Dec 2026 (date not yet announced)	Q3 FY27 results (quarter to October)	Revenue against US$3.15bn +/- 5%; data centre up more than 20% sequentially; gross margin
Early Mar 2027	Q4 FY27 results and FY27 10-K	FY28 guide against US$20bn; ten largest customers' share; custom against interconnect
Through FY28	Custom ramp towards the FY29 target	Whether custom revenue tracks to more than three times FY28 in FY29
Through 2027	1.6T ramp; first scale-up optics revenue	Whether 1.6T links keep a DSP on board; co-packaged run rate towards US$500m

Risks: what would hurt the view

The cycle turns. If the largest cloud companies slow spending, volume falls whatever Marvell's share of each link; the bear case is a 46% fall. **Custom is lumpy.** Programmes are won and lost by generation, the buyers run their own chip teams, and gross margin already slips as custom grows. **Targets are not results.** The fiscal 2031 range is a plan set in a presentation, not guidance, and Marvell had not filed it with the SEC by 8 October. **The translation moves.** If 1.6T or 3.2T links drop the DSP for linear optics, or co-packaged optics spread from scale-up into scale-out, the toll moves to whoever makes the switch chip; Marvell makes switch chips and now co-packaged parts, so it would compete on different ground rather than lose outright.

Few, large buyers. The ten largest customers took 82% of FY26 revenue, up from 81%; one distributor took 44% of revenue in the August quarter, against 34% a year earlier. **Quality of earnings.** Non-GAAP figures exclude stock-based pay of 11.9% of revenue, which has more than doubled in a year.

What would change my mind

Out of the long, into no call: a price above about US$310 on unchanged estimates, which would leave my base case less than 15% above it, or evidence that FY29 earnings will fall below about US$9.92, the level the long needs at today's price, such as Marvell cutting the FY29 custom target back towards US$10bn. **Into more conviction:** the Investor Day targets repeated in an SEC filing or a results release, and data centre revenue in the quarter to October rising more than 20% on the quarter before, as guided.

The call is wrong if the shares close below US$230 by October 2027, or Marvell's fiscal 2028 revenue, the year to January 2028, comes in below US$18 billion.

Conclusion

My piece of 28 September argued that Marvell is paid every time AI chips talk, through the signal processor in each optical plug. The year of results behind it supports that claim: interconnect is the largest part of the data centre business, guided to grow more than 70% this year, and Marvell expects pluggable optics to stay the norm in scale-out networks. Its US$1bn of prepayments buys capacity for the whole business, not optics in particular. The Investor Day raised fiscal 2028 to about US$20bn and kept interconnect the largest line in the fiscal 2031 plan. The price needs only about US$22.5bn of FY29 revenue; I take US$28bn, below management's path, at the peer median multiple. My call: LONG, medium conviction, target US$355; range US$153 to US$517. Medium, because the plan is a presentation, not yet results.

Appendix: forecast assumptions

Assumption	FY27E	FY28E	FY29E	Basis
Revenue, US$bn	12.00	20.00	28.00	Q3 guide midpoint; Q4 3.69 from the US$12bn outlook; FY28 Investor Day outlook; FY29 mine
Gross margin (non-GAAP)	58.4%	58.0%		H2 FY27 guide 57.5 to 58.5%; same range in FY28 (CFO)
Operating expenses, US$bn	2.55	3.40		FY27 guide US$2.55bn; FY28 grows at half the rate of revenue
Operating margin (non-GAAP)	37.1%	41.0%	43.0%	Prior model 38 to 40%; FY31 target model 44 to 46%
Interest and other, US$bn	-0.144	-0.140	-0.120	Q3 guide about US$36m expense a quarter
Tax rate (non-GAAP)	11%	13%	15%	11% in FY27; about 13% guided for FY28; 15% in the FY31 model
Diluted shares, m		930	940	Q3 guide 921m; awards, warrant and Celestial shares net of buybacks
Target multiple			33x	Median of six peers: ALAB 69.9x, LITE 51.5x, COHR 36.0x, CRDO 30.8x, AVGO 20.9x, NVDA 19.8x

Full workings, with live formulas, in the accompanying model, 2026-10-08_Marvell_Model.xlsx.

Sources

Marvell quarterly results releases furnished on Form 8-K (Exhibit 99.1), 3 December 2024 to 27 August 2026, with their comparative columns, for revenue, data centre revenue, non-GAAP margins, EPS, stock-based pay, balance sheet and the Q3 FY27 guide. Marvell Form 10-Q for the quarter to 1 August 2026 (filed 28 August 2026) for shares outstanding, Celestial AI and its contingent consideration, prepayments on supply capacity reservation agreements, capacity reservation arrangements and customer shares. Marvell Forms 10-K for FY26 (filed 11 March 2026) and FY25 (filed 12 March 2025) for ten-largest-customer shares and purchase commitments. Marvell Forms 8-K of 31 March 2026 (Nvidia preferred stock) and 19 August 2026 (Google warrant). Marvell earnings calls of 5 March, 27 May and 27 August 2026. Marvell Investor Day presentation, 6 October 2026, investor.marvell.com, for the FY28 and FY31 revenue targets, data centre and custom targets, the FY31 split and the FY31 target model; transcript of the remarks from stockanalysis.com. Share prices from Nasdaq.com historical data to 5 October and stockanalysis.com for 6 and 7 October; consensus EPS, average target and multiples from stockanalysis.com, retrieved 6 to 8 October 2026. Estimates for FY27 to FY29 and the data centre split are the author's own.

About this note

I wrote this note and built the accompanying model independently. It is not a product of any bank or broker. It tests a claim from my journal piece of 28 September 2026, thephysicallayer.fyi/journal/marvell-collects-the-toll; all notes and their track record are at thephysicallayer.fyi/research. Figures from company filings are labelled as such; derived figures are marked; the forecasts and view are my own estimates and my own view. I hold no position in Marvell. Personal research, not investment advice.

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